7.11 PreQuiz: Logarithmic and Exponential Functions — Markscheme

1. [Maximum mark: 6]

   Noah invests \$3200 at 7.2 % per annum, compounded quarterly.

   (a) Quarterly rate $= \dfrac{0.072}{4} = 0.018.$ **M1**

   $V(t) = 3200(1.018)^{4t}$ **A1**

   (b) Since $(1.018)^{4t} = e^{4t\ln(1.018)}$, **M1**

   $V(t) = 3200e^{kt}$ where $k = 4\ln(1.018)$. **A1**

   (c) Need the smallest integer n with $3200(1.018)^{4n} > 5000$. **M1**

   $n > 6.25\ldots$, so the smallest full-year value is $n = 7$ years. **A1**

   Alternative method (table): show both adjacent rows $3200(1.018)^{24} = 4910.48$ *(at* $n = 6$*) and* $3200(1.018)^{28} = 5273.18$ *(at* $n = 7$*), then state* $n = 7$*. A single-row table earns full marks but should receive a feedback note recommending the adjacent row to justify the crossover.*

2. [Maximum mark: 6]

$U(t) = 9000e^{kt}$ and $U(3) = 18\,500$.

(a) $9000e^{3k} = 18\,500$, so $e^{3k} = \dfrac{18500}{9000}$. **M1**

$k = \dfrac{1}{3}\ln\left(\dfrac{18500}{9000}\right) = 0.2401\ldots$ **A1**

(b) $U(7) = 9000e^{7k}$, using their value of k. **M1**

$U(7) = 48\,351.58\ldots$, so $48\,352$ users. **A1**

Accept 48 290 *(obtained by substituting rounded* $k = 0.240$*) with a feedback note: do not round intermediate values; carry full precision through and round only at the end.*

(c) Set $9000e^{kt} = 50\,000$. **M1**

$t = \dfrac{\ln(50000/9000)}{k} = 7.14$ years. **A1**

Accept 7.15 *(obtained by substituting rounded* $k = 0.240$*) with the same feedback note about not rounding intermediate values.*